

Data Scaling and Working with Dataframes

When training models on data, often times it is useful to scale input features, so they all have a similar range of values. There are multiple ways to apply this scaling, and you will build functions to perform some of these. You will then apply this on some data.

1. Data Normalization

A common method to scale a data set feature is to **normalize** it. This puts all of the features values on a scale between 0 and 1. Given a set of values $X_1, X_2, \dots, X_n$, each corresponding normalized value is calculated using the following formula.

$$\text{normalized } X_i = \frac{X_i - \min\{X_1, X_2, \dots, X_n\}}{\max\{X_1, X_2, \dots, X_n\} - \min\{X_1, X_2, \dots, X_n\}}$$

Build a Python function that takes in a vector (array) and normalizes it.

2. Data Standardization

Another common method to scale a data set feature is to **standardize** it. This calculates the z -score of a feature value. Given a set of values $X_1, X_2, \dots, X_n$, with sample mean $\bar{X}$ and sample standard deviation s_X , then each corresponding standardized value is calculated using the following formula.

$$\text{standardized } X_i = \frac{X_i - \bar{X}}{s_X}$$

Build a Python function that takes in a vector (array) and standardizes it.

3. Working with a Dataframe

In this problem, you will be working with the data set **calif_housing_data.csv**. This data is a modified version of the data set from <https://www.kaggle.com/camnugent/california-housing-prices?select=housing.csv>. This data set has housing information on various California block neighborhoods. There are five columns in the data, which include the median age (in years) of houses on the block, the total number of bedrooms on the block, the total number of households on the block, the median income on the block, and the median house value on the block. Now, suppose you are building a model to predict the median house value. Use Python code to answer the following questions

- How many rows does this data set have?
- What is the target vector for your model?
- Create a new feature by taking the total bedrooms divided by the number of households. What does this new feature represent?
- Now, create a new data frame that has three features: the median age, median income, and the new feature created in part (c).
- Take the data frame created in part (d) and apply data standardization to the features.